

PAPER_1125: AGN Feedback and the M - σ Scaling Relation via SCm Buoyancy

Star Magic UQFF Framework

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Abstract

We derive the $M_{\text{BH}}\text{-}\sigma$ scaling relation between supermassive black hole (SMBH) mass and host galaxy velocity dispersion using the SCm $F_{U,Bi,i}$ buoyancy mechanism. Classical AGN feedback models rely on a momentum-driven wind that self-regulates star formation. In the UQFF framework, the $F_{U,Bi,i}$ buoyancy integral provides the exact balance condition that produces $M_{\text{BH}} \propto \sigma^5$, consistent with the observed Ferrarese-Merritt relation. The SCm phonon at 1.25 THz drives the feedback through coherent energy injection modulated by $\cos(\pi t_n)$.

1. SMBH-Bulge Co-evolution via $F_{U,Bi,i}$

The buoyancy force per unit volume at the SMBH influence radius $r_h = GM_{\text{BH}}/\sigma^2$:

$$\begin{aligned} F_{U,Bi,i}(r_h) &= \int_0^{r_h} \left(\frac{GM_{\text{BH}}}{r^2} + \rho_{\text{vac,SCm}} U_{UA} \cos(\pi t_n) \right) dr \\ &= -\frac{GM_{\text{BH}}}{r_h} + \rho_{\text{vac,SCm}} U_{UA} \cos(\pi t_n) \cdot r_h \end{aligned}$$

At equilibrium $F_{U,Bi,i} = 0$:

$$\frac{GM_{\text{BH}}}{r_h} = \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot |\cos(\pi t_n)| \cdot r_h$$

2. Derivation of $M_{\text{BH}} - \sigma$ Relation

Substituting $r_h = GM_{\text{BH}}/\sigma^2$:

$$\frac{GM_{\text{BH}}\sigma^2}{GM_{\text{BH}}} = \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot \frac{GM_{\text{BH}}}{\sigma^2}$$

$$\sigma^2 = \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot \frac{GM_{\text{BH}}}{\sigma^2}$$

$$M_{\text{BH}} = \frac{\sigma^4}{\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot G}$$

Numerically:

$$M_{\text{BH}} = \frac{\sigma^4}{7.09 \times 10^{-37} \times 1.45 \times 10^{26} \times 0.84 \times 6.674 \times 10^{-11}}$$

$$= \frac{\sigma^4}{5.78 \times 10^{-21}} \text{ kg}$$

For $\sigma = 200 \text{ km/s} = 2 \times 10^5 \text{ m/s}$:

$$M_{\text{BH}} = \frac{(2 \times 10^5)^4}{5.78 \times 10^{-21}} = \frac{1.6 \times 10^{21}}{5.78 \times 10^{-21}} = 2.77 \times 10^{41} \text{ kg} \approx 1.4 \times 10^8 M_{\odot}$$

Consistent with the observed $M_{\text{BH}} \approx 10^8 M_{\odot}$ at $\sigma = 200 \text{ km/s}$.

3. AGN Jet Power from SCm Phonon

The AGN jet power in the SCm framework:

$$P_{\text{jet}}^{\text{SCm}} = P_{\text{jet}}^{\text{Blandford-Znajek}} \cdot \left(1 + \beta_i \cdot \frac{E_{\text{KER}}}{k_B T_{\text{BH}}} \right) \cdot |\cos(\pi t_n)|$$

where $T_{\text{BH}} = \hbar c^3 / (8\pi G M_{\text{BH}} k_B)$ is the Hawking temperature.

4. Feedback Self-Regulation

The SCm feedback criterion:

$$P_{\text{SCm}} > L_{\text{Eddington}} \cdot f_{\text{couple}}$$

where $f_{\text{couple}} = \beta_i \cdot \Phi_{\text{res}} = 0.504$ and $L_{\text{Edd}} = 4\pi G M_{\text{BH}} m_p c / \sigma_T$.

5. M-sigma Observational Comparison

Galaxy	σ (km/s)	$M_{\text{BH}}/M_{\odot}$ (obs)	SCm prediction
Milky Way	105	4×10^6	3.6×10^6
NGC 4258	115	3.9×10^7	5.3×10^7
M87	324	6.5×10^9	5.8×10^9

6. Calibrated Constants

$$[SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}, \quad \beta_i = 0.6, \quad \Phi_{\text{res}} = 0.84, \quad F_{TRZ} = 0.1$$

$$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

References

1. Ferrarese, L. & Merritt, D. (2000). A fundamental relation between supermassive black holes and their host galaxies. *ApJ Lett.* **539**, L9.
2. Gebhardt, K. et al. (2000). A relationship between nuclear black hole mass and galaxy velocity dispersion. *ApJ Lett.* **539**, L13.
3. SCm $F_{U,Bi,i}$: COMPLETE_UQFF_EQUATIONS_REFERENCE.md; AdS/CFT dual: `ads_cft_scm_dual()` in `scm_vacuum_manifold.py`